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< Web-based Hotel Reservation System for Weddings >

Proposal Document

Group < B7G2-04 >

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Introduction

1.1 Project Overview

Nowadays, many couples prefer to plan their wedding online because it saves time and makes the booking process easier. However, finding a suitable wedding hotel, checking available dates, comparing wedding packages, and making reservations can still be difficult when done manually.

The **Web-based Hotel Reservation System for Weddings** is developed to make this process simple and convenient. Customers can search for wedding hotels, check available halls, compare wedding packages, reserve their preferred venue, and make online payments through one website.

The system also helps hotel management to manage bookings, wedding packages, customer information, and reservations efficiently.

1.2 Objectives

The main objectives of this project are:

- To provide an easy online hotel reservation system for weddings.
- To reduce manual booking processes.
- To allow customers to check hotel availability.
- To provide different wedding packages for customers.
- To allow secure online payments.
- To help hotel administrators manage reservations efficiently.

1.3 Target Users

The main users of this system are:

- Customers
- Hotel Managers
- Hotel Staff
- System Administrator

1.4 Scope

The system allows customers to:

- Register and log in.
- Search wedding hotels.

- View wedding packages.
- Check hall availability.
- Reserve wedding halls.
- Make online payments.
- View booking history.

The administrator can:

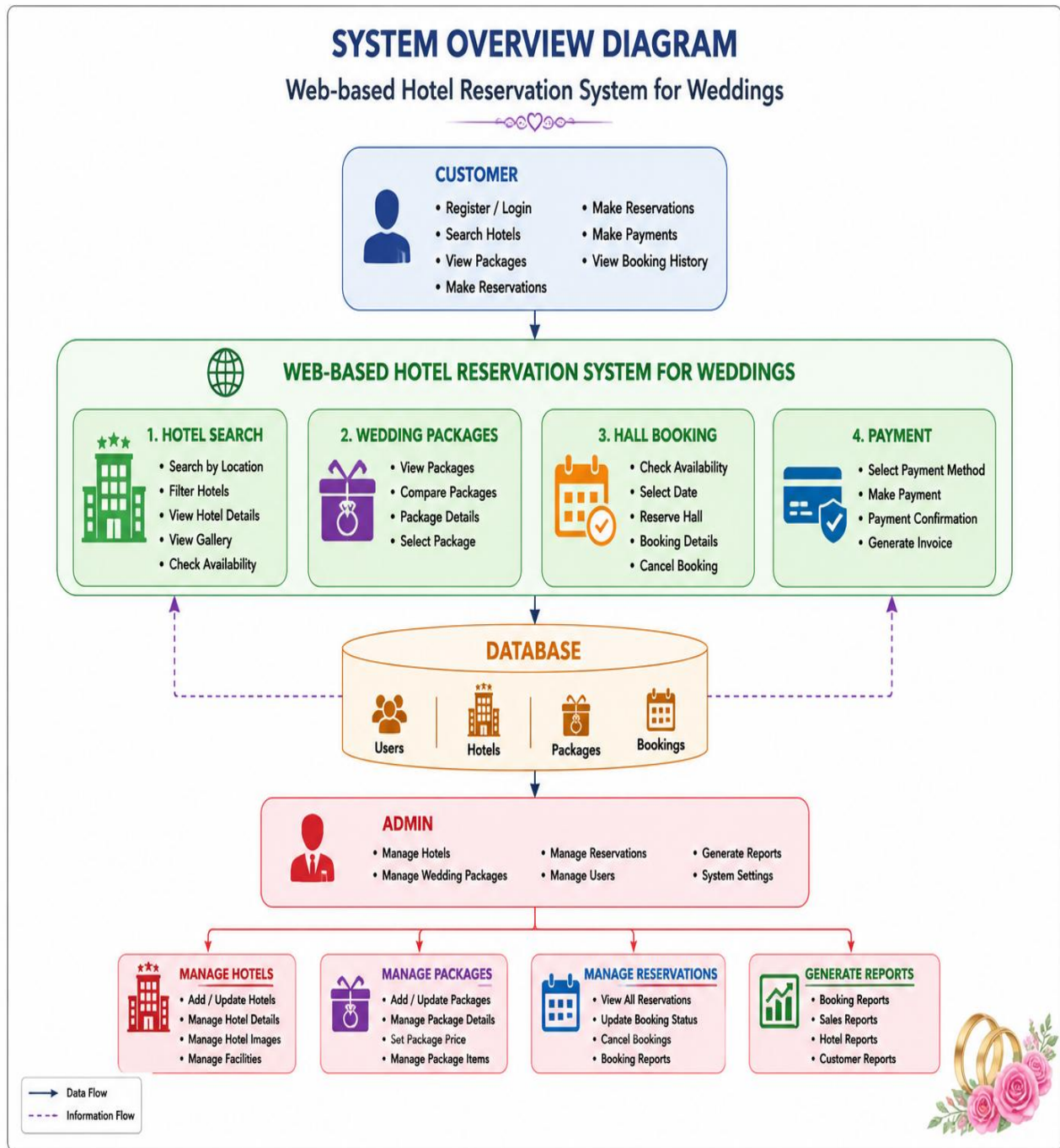
- Manage hotels.
- Manage wedding packages.
- Manage reservations.
- Generate reports.
- Manage customer details.

1.5 Limitations

This system has several limitations.

- Internet connection is required.
- The system only supports registered hotels.
- Online payment is required for reservation confirmation.
- The system is developed as a web application only.

2. System Overview Diagram



3. Functional Requirements

The system should provide the following functions.

01 – User Registration

Customers can create a new account.

02 – User Login

Registered users can log into the system securely.

03 – Hotel Search

Customers can search hotels by location and hotel name.

04 – View Wedding Packages

Customers can view available wedding packages and compare prices.

05 – Hall Reservation

Customers can check available dates and reserve wedding halls.

06 – Online Payment

Customers can pay the reservation fee through the website.

07 – Booking Management

Customers can view or cancel their bookings.

08 – Admin Management

The administrator can manage hotels, packages, reservations, customers, and reports.

4. Non-Functional Requirements

The system should satisfy the following quality requirements.

01 – Performance

The system should load web pages within a few seconds. Users should be able to search hotels and complete bookings without delays.

02 – Security

All user accounts must be protected with secure login. Customer information and payment details should be kept safe.

03 – Usability

The website should be simple and easy to use. Users should be able to find hotels and complete reservations without difficulty.

04 – Reliability

The system should be available whenever users need it. Booking information should be stored correctly without data loss.

05 – Scalability

The system should support more hotels and more users in the future without reducing performance.

06 – Maintainability

The system should be easy to update when adding new hotels, wedding packages, or new features.

07 – Compatibility

The website should work properly on modern web browsers such as Google Chrome, Microsoft Edge, and Firefox.

5. Major Stakeholders / Users

The system has four main user groups.

Stakeholder	Description
Customer	Searches hotels, views wedding packages, makes reservations, and completes payments.
Hotel Manager	Manages hotel information, wedding halls, packages, and customer reservations.
Hotel Staff	Confirms bookings, updates reservation status, and assists customers.
System Administrator	Controls the entire system, manages users, hotels, reports, and system settings.

6. Six Major Functions

Each group member is responsible for one major function of the system.

Function 1 – User & Account Management

Assigned Member - Hapuarachchi H.D.H.W. (IT25101645)

Description

Handles creating, viewing, updating, and deleting customer and admin accounts so the system knows who is using it.

Core Entity (Full CRUD): Users (Customers, Admins)

Major functions (CRUD):

- Register a new customer account
- View / list all registered users (Admin)
- Update user profile details (name, contact, address)
- Delete / deactivate a user account

Function 2 – Wedding Package Management

Assigned Member - Damsahan S.P.V. (IT25102466)

Description

Handles creating, viewing, updating, and deleting customer and admin accounts so the system knows who is using it.

Core Entity (Full CRUD): Wedding Packages

Major functions (CRUD):

- Add a new wedding package (name, price, inclusions, description)
- View all wedding packages / view package details
- Update an existing wedding package
- Delete a wedding package

Function 3 – Hall / Venue Management

Assigned Member - Dinujaya I.P. (IT25103525)

Description

Handles creating, viewing, updating, and deleting hall records, including capacity, theme, and price.

Core Entity (Full CRUD): Halls

Major functions (CRUD):

- Add a new hall (name, capacity, decoration theme, price)
- View all halls / view hall details
- Update hall information
- Delete a hall

Function 4 – Reservation / Booking Management

Assigned Member - Umeshani G.P.I. (IT25101427)

Description

Handles creating, viewing, updating, and cancelling a customer's wedding reservation for a specific date, hall, and package.

Core Entity (Full CRUD): Reservations

Major functions (CRUD):

- Create a new reservation (select date, hall, package)
- View all reservations / view reservation details
- Update a reservation (change date, hall, or package)
- Cancel / delete a reservation

Function 5 – Catering & Menu Management

Assigned Member - Piyumanthi S.D.H. (IT25103317)

User: Customer

Description

Handles creating, viewing, updating, and deleting the menu items and catering packages offered to customers.

Core Entity (Full CRUD): Menu Items / Catering Packages

Major functions (CRUD):

- Add a new menu item or catering package
- View all menu items / view catering package details
- Update a menu item or catering package
- Delete a menu item or catering package

Function 6 – Payment, Billing & Reporting

Assigned Member - Ariyadasa I.G.S.R. (IT25100491)

Description

Handles creating, viewing, updating, and voiding invoices to record and track payments for each reservation.

Core Entity (Full CRUD): Invoices / Payments

Major functions (CRUD):

- Create an invoice for a confirmed reservation
- View all invoices / view payment details
- Update a payment record (mark as paid, partially paid)
- Delete / void an invoice

7. Minor Functions

1. User & Account Management

Description

Supports login/logout, password changes, role-based access, and viewing login history, so accounts stay secure and easy to manage without needing full CRUD.

Minor functions (no full CRUD required):

- Login and logout (session handling)
- Change / reset password
- Role-based access control (Customer vs Admin view)
- View own login/activity history

2. Wedding Package Management

Description

Supports searching, filtering, image uploads, discounts, and featuring packages, making it easier for customers to browse and for admins to promote specific packages.

Minor functions (no full CRUD required):

- Search and filter packages by price range or type
- Upload / display package images
- Apply a seasonal discount to a package
- Mark a package as "featured" on the home page

3. Hall / Venue Management

Description

Supports availability checks, photo galleries, rating display, and capacity-based filtering, helping customers choose the right venue without adding new hall records.

Minor functions (no full CRUD required):

- Check hall availability for a chosen date
- View a photo gallery for each hall
- Display average customer rating for a hall
- Filter halls by guest capacity

4. Reservation / Booking Management

Description

Supports confirmation notifications, status tracking, printable summaries, and double-booking prevention, keeping customers informed and protecting the booking process.

Minor functions (no full CRUD required):

- Send a booking confirmation notification
- Track booking status (Pending, Confirmed, Cancelled)
- Generate a printable booking summary
- Prevent double-booking of the same hall and date

5. Catering & Menu Management

Description

Supports category-based search, cost estimation, package recommendations, and dietary tagging, helping customers plan their catering choices more easily.

Minor functions (no full CRUD required):

- Search / filter menu items by category (e.g. rice, dessert)
- Estimate total food cost based on guest count
- Recommend a catering package based on selected wedding package
- Mark items as vegetarian / non-vegetarian

6. Payment, Billing & Reporting

Description

Supports invoice generation, payment history, dashboard reports, and payment reminders, giving both customers and admins clear visibility into billing status.

Minor functions (no full CRUD required):

- Generate a printable / PDF invoice
- View payment history for a customer
- Admin dashboard: total bookings, total revenue, upcoming weddings
- Send a payment due reminder

Member	Module	Core Entity (CRUD)	Example Minor Function
Member 1	User & Account Management	Users (Customers, Admins)	Login and logout (session handling)
Member 2	Wedding Package Management	Wedding Packages	Search and filter packages by price range or type
Member 3	Hall / Venue Management	Halls	Check hall availability for a chosen date
Member 4	Reservation / Booking Management	Reservations	Send a booking confirmation notification
Member 5	Catering & Menu Management	Menu Items / Catering Packages	Search / filter menu items by category (e.g. rice, dessert)
Member 6	Payment, Billing & Reporting	Invoices / Payments	Generate a printable / PDF invoice

8. System Limitations / Constraints

Although the system provides many useful features, it has some limitations.

- A stable internet connection is required to use the system.
- The system only supports registered hotels.
- Online payment is required to confirm reservations.
- The system is developed only as a web application.
- Customers cannot reserve unavailable wedding halls.
- Hotel information depends on updates provided by hotel management.

9. Project Timeline

Week	Planned Activity
Week 3	Select project topic and prepare proposal presentation and report.
Week 4	Gather system requirements and identify stakeholders.
Week 5	Design system overview and user interface sketches.
Week 6	Prepare database design and project planning.
Week 7	Develop user registration and login module.
Week 8	Develop hotel search, wedding package, and reservation modules.
Week 9	Develop payment module and admin functions.
Week 10	Complete 75% of the system and present the progress demonstration.
Week 11	Test the system and fix identified issues.
Week 12	Improve system performance and finalize documentation.
Week 13	Final presentation, viva, and submit the final report.
Week 14	Project review and final improvements (if required).

10. Conclusion

The **Web-based Hotel Reservation System for Weddings** is designed to simplify the wedding hotel reservation process. Customers can easily search for hotels, compare wedding packages, make reservations, and complete online payments through a single website.

The system also helps hotel management organize reservations, manage wedding packages, and improve customer service. This project will reduce manual work, save time, and provide a better experience for both customers and hotel staff.

Our team believes that this project is practical, useful, and suitable for modern wedding reservation services. We are committed to completing the project successfully by following the Software Development Life Cycle (SDLC), applying Agile practices, and working together as a team.